

Randomized Quantiles of Weighted Bernoulli Sums

An addendum to the computation note

October 7, 2026

Abstract

For a discrete distribution, a deterministic quantile generally leaves unused probability at its boundary. Independent randomization between two adjacent support points removes this discrepancy when the distribution can be computed exactly. This addendum gives the one-sided construction and the original two-sided construction for fair weighted Bernoulli sums, proves their probability identities, and explains the special treatment of an atom at the center. It also specifies a conservative fallback and its different output interpretation. Both constructions use the existing single public function, `qweighted_bernoulli()`.

1 Model, interface, and probability conventions

Let

$$G = \sum_{i=1}^B w_i \xi_i, \quad \xi_i \sim \text{Ber}(\pi_i) \text{ independently}, \quad w_i \geq 0, \quad 0 \leq \pi_i \leq 1,$$

and write $F(x) = \mathbb{P}(G \leq x)$. The weights and success probabilities are specified inputs. All probability identities involving a randomized cutoff include a fresh uniform variable $V \sim U(0, 1)$, independent of G and of any data or tie-breaking variables used elsewhere in the procedure.

The package retains one exported function. Three arguments are appended to the existing interface, preserving earlier positional calls:

```
qweighted_bernoulli(  
  p, weights, prob = 0.5, max_length = 25,  
  lower.tail = TRUE, log.p = FALSE, details = FALSE,  
  control = list(),  
  randomized = FALSE, two.sided = FALSE, draw = TRUE  
)
```

The default `randomized = FALSE` retains deterministic quantile computation. For `randomized = TRUE`, the function always returns a list containing `c_plus`, `c_minus`, `tau`, `P_plus`, `P_minus`, `exact`, and `kappa`. The field `tau` is always the probability of selecting the larger cutoff `c_plus`. Setting `draw = FALSE` returns the mixing distribution with `kappa = NULL`; it does not generate a draw.

The argument `p` retains the usual probability convention. With `lower.tail = FALSE`, it specifies the tail level α directly. With `lower.tail = TRUE`, it specifies $1 - \alpha$. Setting `log.p = TRUE` supplies the logarithm of the corresponding probability. The statements below use $0 < \alpha < 1$.

2 Exact one-sided randomization

Put $p_0 = 1 - \alpha$ and define

$$c_+ = \min\{x \in \text{supp}(G) : F(x) \geq p_0\}, \quad c_- = \max\{x \in \text{supp}(G) : x < c_+\}.$$

If the latter set is empty, use the sentinel $c_- = -\infty$. This convention also works when deterministic summands shift the minimum of the support away from zero. Define

$$P_+ = \mathbb{P}(G > c_+), \quad P_- = \mathbb{P}(G > c_-), \quad \tau = \frac{P_- - \alpha}{P_- - P_+}. \quad (1)$$

Minimality of c_+ gives $P_+ \leq \alpha < P_-$ and hence $0 < \tau \leq 1$. The randomized cutoff is

$$\kappa = \begin{cases} c_+, & V \leq \tau, \\ c_-, & V > \tau. \end{cases} \quad (2)$$

Proposition 2.1. *The cutoff in (2) satisfies $\mathbb{P}(G > \kappa) = \alpha$, or equivalently $\mathbb{P}(G \leq \kappa) = 1 - \alpha$.*

Proof. Independence permits conditioning on the selected cutoff, giving

$$\mathbb{P}(G > \kappa) = \tau P_+ + (1 - \tau) P_- = P_- - \tau(P_- - P_+) = \alpha.$$

The complementary identity follows because the event uses a strict upper tail. No continuity assumption is needed. $\square$

Equivalently,

$$\tau = \frac{p_0 - \mathbb{P}(G < c_+)}{\mathbb{P}(G = c_+)}.$$

This identity explains the need for the atom at the quantile. Merely returning the ordinary quantile does not provide the randomization. For a constant $G = g_0$, the construction gives $c_+ = g_0$, $c_- = -\infty$, and $\tau = p_0$. The natural one-sided endpoint extensions are a cutoff of $-\infty$ at $p_0 = 0$ and the maximum support point at $p_0 = 1$.

```
one <- qweighted_bernoulli(
  p = 0.05, weights = c(1, 2, 4),
  prob = c(0.2, 0.5, 0.8), lower.tail = FALSE,
  randomized = TRUE, draw = FALSE
)
one[c("c_plus", "c_minus", "tau", "P_plus", "P_minus")]

# In the exact branch, this is the requested strict-tail level.
one$tau * one$P_plus + (1 - one$tau) * one$P_minus
```

Here `P_plus` and `P_minus` are the strict upper-tail probabilities in (1). In particular, the probability associated with the larger cutoff is the smaller tail probability.

3 The original fair two-sided cutoff

Now assume $\pi_i = 1/2$ for every positive weight, and let $N = \sum_i w_i$. Complementing all Bernoulli variables gives $G \stackrel{d}{=} N - G$. Define

$$\mathcal{S} = \{-1\} \cup (\text{supp}(G) \cap [0, N/2]),$$

and, for $c \in \mathcal{S} \setminus \{-1\}$,

$$P(c) = \mathbb{P}\left(\left|G - \frac{N}{2}\right| \geq \frac{N}{2} - c\right), \quad P(-1) = 0.$$

The value -1 is a sentinel below the nonnegative support. It is appropriate even for noninteger weights: an increment of one between support points is neither assumed nor used.

Proposition 3.1. *Let*

$$c_+ = \min\{c \in \mathcal{S} : P(c) \geq \alpha\}, \quad c_- = \max\{c \in \mathcal{S} : c < c_+\}.$$

Then $c_+ = q_G(\alpha/2)$, where q_G is the ordinary leftmost distributional quantile. Write $F_- = \mathbb{P}(G < c_+)$. The probabilities at the two cutoffs are

$$P_- = 2F_-, \quad P_+ = \begin{cases} 2F(c_+), & c_+ < N/2, \\ 1, & c_+ = N/2. \end{cases} \quad (3)$$

With

$$\tau = \frac{\alpha - P_-}{P_+ - P_-} \quad (4)$$

and the independent mixture (2),

$$\mathbb{P}\left(\left|G - \frac{N}{2}\right| \geq \frac{N}{2} - \kappa\right) = \alpha.$$

Proof. For $c < N/2$, the events $\{G \leq c\}$ and $\{G \geq N - c\}$ are disjoint. Symmetry therefore gives $P(c) = 2F(c)$. At $c = N/2$, their union is the whole space, so $P(N/2) = 1$. Symmetry also gives $F(N/2) \geq 1/2$. Since $\alpha/2 < 1/2$, the quantile $q_G(\alpha/2)$ is at most $N/2$ and is the first support point at which $P(c) \geq \alpha$. Its predecessor is strictly below $N/2$, and $F(c_-) = \mathbb{P}(G < c_+)$, including the sentinel case. These facts prove (3). Minimality gives $P_- < \alpha \leq P_+$, so the mixing probability is in $(0, 1]$. Conditioning on the independent mixture yields $(1 - \tau)P_- + \tau P_+ = \alpha$, as claimed. $\square$

The center atom. If $c_+ = N/2$, the jump $P_+ - P_-$ equals $\mathbb{P}(G = N/2)$, not twice that atom. Indeed, symmetry gives $2F_- = 1 - \mathbb{P}(G = N/2)$. Replacing P_+ by $2F(c_+)$ would double-count the center and use the wrong randomization.

For $w = (1, 1)$ and $\alpha = 0.75$, the law of G has masses $1/4, 1/2, 1/4$ at $0, 1, 2$. Thus

$$c_- = 0, \quad c_+ = 1, \quad P_- = \frac{1}{2}, \quad P_+ = 1, \quad \tau = \frac{1}{2}.$$

For all-zero weights, $G = N = 0$ and the same definitions give $c_- = -1$, $c_+ = 0$, and $\tau = \alpha$.

```
center <- qweighted_bernoulli(
  p = 0.75, weights = c(1, 1), lower.tail = FALSE,
  randomized = TRUE, two.sided = TRUE, draw = FALSE
)
center[c("c_plus", "c_minus", "tau", "P_plus", "P_minus")]
# c_plus = 1, c_minus = 0, tau = 0.5
# P_plus = 1, P_minus = 0.5

cut <- qweighted_bernoulli(
  p = 0.05, weights = 1:20, lower.tail = FALSE,
  randomized = TRUE, two.sided = TRUE, draw = FALSE
)
cut$tau * cut$P_plus + (1 - cut$tau) * cut$P_minus
```

The two-sided mode uses the probabilities in (3); the larger cutoff now has the larger event probability. This explains the different orientations of (1) and (4), although `tau` selects the larger cutoff in both modes.

4 Obtaining the bracket without full enumeration

The exact engines and the rule `max_length = 25` are described in the main note [1]. Randomization uses the same exact law and adds the predecessor and the boundary probabilities. It does not require a new public function or full enumeration of all 2^B Bernoulli outcomes.

In a meet-in-the-middle calculation, write $G = A + D$, where A and D are independent half-sums with sorted supports a_i and d_j and masses r_i and s_j . At a candidate x , determine

$$j_i^{\leq}(x) = \max\{j : a_i + d_j \leq x\}, \quad j_i^{<}(x) = \max\{j : a_i + d_j < x\},$$

using zero when the respective set is empty. With prefix masses $S_j = \sum_{k \leq j} s_k$ and $S_0 = 0$, the exact queries are

$$F(x) = \sum_i r_i S_{j_i^{\leq}(x)}, \quad \mathbb{P}(G < x) = \sum_i r_i S_{j_i^{<}(x)}.$$

The actual predecessor of x is

$$\max_{i: j_i^{<}(x) > 0} (a_i + d_{j_i^{<}(x)}),$$

or the appropriate sentinel if the index set is empty. These same strict and non-strict queries already verify the jump containing an exact quantile. Reusing them avoids enumerating the Cartesian product of the two half-supports.

The repeated-class engine applies the same calculation after replacing equal weight–probability pairs by binomial counts. Binomial and Wilcoxon supports have consecutive lattice coordinates. A general lattice distribution can have gaps, so its predecessor must be the previous reachable coordinate. Subtracting a unit or an arbitrary numerical tolerance from an exact quantile is not a general substitute for this search.

Numerical meaning of exact. The probability identities above are mathematical identities. As in the main note, `exact = TRUE` describes an exact finite-law algorithm implemented in double precision; it does not assert interval-certified arithmetic. Direct small-tail evaluation, logarithmic probabilities, and stable differences avoid unnecessary cancellation. On a recognized lattice, integer coordinates also determine whether the quantile is the center. Arbitrary tolerance-based merging of nearby atoms is not justified by the theory.

5 A conservative fallback with a stated guarantee

If the exact law is unaffordable, the existing algorithm computes the applicable conservative candidates under its work limits and chooses their minimum [1]. The selected deterministic bound is chosen before any random draw. A bound alone does not identify the quantile atom, and its upper-tail certificate cannot be inserted into an exact interpolation formula as though it were the true tail probability.

One-sided mode

Suppose the selected threshold q has a certificate

$$\mathbb{P}(G > q) \leq b \leq \alpha.$$

The fallback reports the degenerate mixture $c_+ = c_- = q$ and $\tau = 1$. It consequently satisfies $\mathbb{P}(G > \kappa) \leq \alpha$. The fields `P_plus` and `P_minus` report bounds on the event probabilities, rather than the event probabilities themselves, and `exact = FALSE` identifies this case. The two reported cutoffs do not assert that the true quantile has been bracketed by adjacent support points.

The original two-sided mode

Suppose the selected upper quantile bound at level $\alpha/2$ has a certificate

$$\mathbb{P}(G > q) \leq b \leq \alpha/2. \quad (5)$$

Choose an actual support value c satisfying

$$0 \leq c \leq N/2, \quad c < N - q, \quad (6)$$

or choose the sentinel $c = -1$ if no such value is obtained.

Proposition 5.1. *Every cutoff chosen in this way satisfies $P(c) \leq \alpha$.*

Proof. The sentinel has $P(-1) = 0$. Otherwise, symmetry and the strict inequality in (6) imply

$$F(c) \leq \mathbb{P}(G < N - q) = \mathbb{P}(N - G > q) = \mathbb{P}(G > q) \leq b.$$

In fact $c < N/2$: if $c = N/2$, then $F(c) \geq 1/2$ by symmetry, contradicting $b \leq \alpha/2 < 1/2$. Thus $P(c) = 2F(c) \leq 2b \leq \alpha$. $\square$

A deterministic greedy subset search gives an inexpensive support value. Start at zero when it satisfies (6); traverse the positive weights and add a weight only when the resulting sum still satisfies that condition. Run the search in ascending and descending order and retain the larger feasible result. Each result is a sum of a subset of the original weights, so it belongs to $\text{supp}(G)$. Optimal subset packing is unnecessary for the guarantee: the strict budget and actual support membership suffice. The sentinel remains available when the budget is nonpositive. Numerical comparisons retain the precision qualification stated above.

The fallback reports $c_+ = c_- = c$, $\tau = 1$, and `exact = FALSE`. Its `P_plus` and `P_minus` are certified bounds, at most $2b$ (zero for the sentinel). These equal endpoints are a degenerate conservative cutoff, not the original exact bracket. The algorithm does not claim equality with α in this branch.

The strict condition $c < N - q$ is essential. A strict upper-tail bound controls $\mathbb{P}(G < N - q)$ by symmetry; setting $c = N - q$ can add a boundary atom that the certificate does not cover.

6 Drawing and interpreting the result

To obtain the mixing distribution without consuming a random draw, use `draw = FALSE`. To obtain the draw as well, use the default `draw = TRUE`:

```
set.seed(20261007)
out <- qweighted_bernoulli(
  p = 0.05, weights = 1:20, lower.tail = FALSE,
  randomized = TRUE, two.sided = TRUE
)
out$kappa
out$exact

# The returned mixing distribution can also be sampled explicitly.
law <- qweighted_bernoulli(
  p = 0.05, weights = 1:20, lower.tail = FALSE,
  randomized = TRUE, two.sided = TRUE, draw = FALSE
)
v <- runif(length(law$tau))
kappa <- ifelse(log(v) < law$log_tau, law$c_plus, law$c_minus)
```

The explicit draw must be independent of the variables to which the cutoff will be applied. In particular, a tie-breaking uniform from an earlier construction must not be reused. The mathematical assertions concern the joint probability over G and this independent randomization; conditional on a realized cutoff, its individual event probability need not equal α .

R's pseudorandom generators have finite resolution; most supplied uniform generators produce at most 2^{32} distinct values [2]. Thus the ideal-uniform probability identities are subject to both arithmetic and simulation discretization. In particular, a single uniform comparison cannot realize every extremely small mixing probability exactly. For such levels, `draw = FALSE` preserves the analytical mixture and its logarithmic weights for a separate sampling scheme. Standard R seeding controls reproducibility; the function does not reset the seed.

Check `exact` before interpreting the probability fields. In an exact row, the weighted average of `P_plus` and `P_minus` equals the requested level, subject to ordinary numerical precision. In a conservative row, those fields are bounds and the weighted average certifies an inequality. All candidate selection is deterministic in the supplied weights, probabilities, levels, and work limits. Taking the minimum of separately randomized upper thresholds is not covered by the guarantee: even two individually valid randomized thresholds can produce an invalid minimum.

References

- [1] *Computing Quantiles of Weighted Bernoulli Sums: Exact Algorithms, Conservative Bounds, and Reproducible R Code* (2026). Technical note accompanying the `weightedBernoulli` package, October 7, 2026. The exact computation engines, concentration inequalities, work limits, and numerical comparisons are documented there. The note is included as `weighted_bernoulli_quantiles.pdf` in the package documentation. This addendum supplies the integrated randomized interface and the probability identities needed to interpret it.
- [2] R Core Team. *Random Number Generation* and *The Uniform Distribution*. R documentation, accessed October 7, 2026. <https://stat.ethz.ch/R-manual/R-devel/library/base/html/Random.html>; <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/Uniform.html>.